

Web Programming Quiz 2 Report

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Github Link :

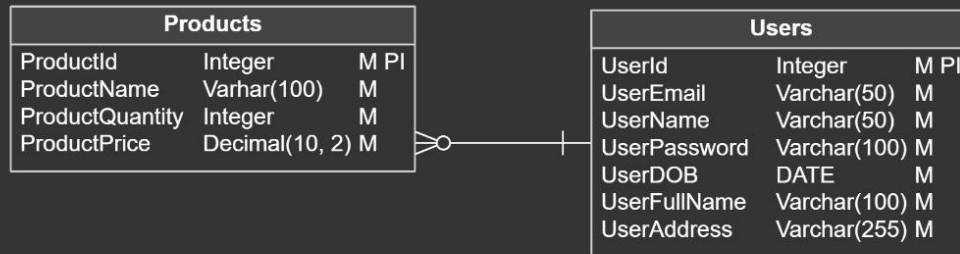
<https://github.com/juangmaulana/Quiz2WebPro>

Introduction

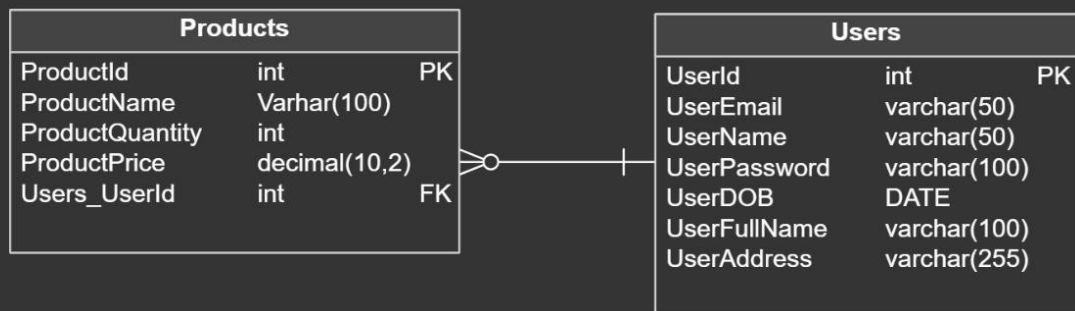
Storify is a website that helps stores or warehouses to display and organize their products. It enables store managers, owners, and staff to add, delete, show, and edit their products. This website uses Jakarta Server Pages (JSP) as its front end interface. For the back end, we are using MySQL for the database and Java Servlets for the code. By combining those features, Storify made database organizing easier and nicer with an interactive and simple user interface.

This report aims to provide a detailed overview of Storify, its functionality, documentation, and an explanation of the code. This web application not only demonstrates CRUD (Create, Read, Update, Delete) operations but highlights the importance of modern database management with an interactive and enjoyable user interface. Through this, Storify ensures data consistency and improves productivity for businesses.

Conceptual Data Model (CDM)

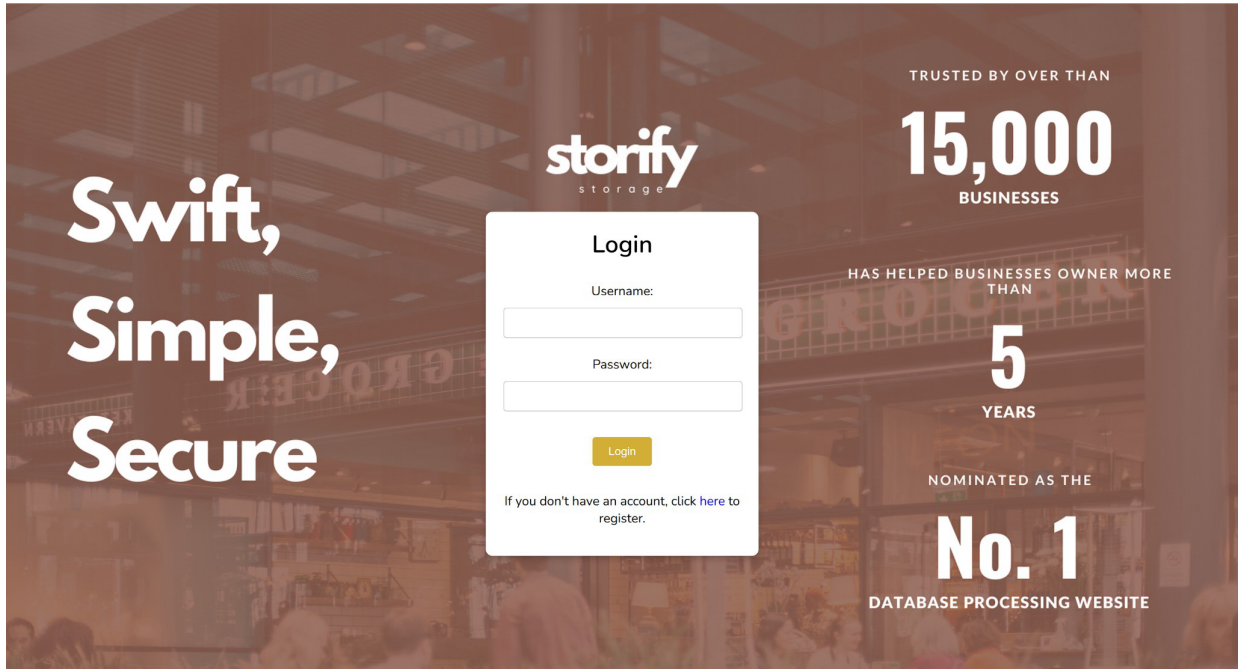


Physical Data Model (PDM)

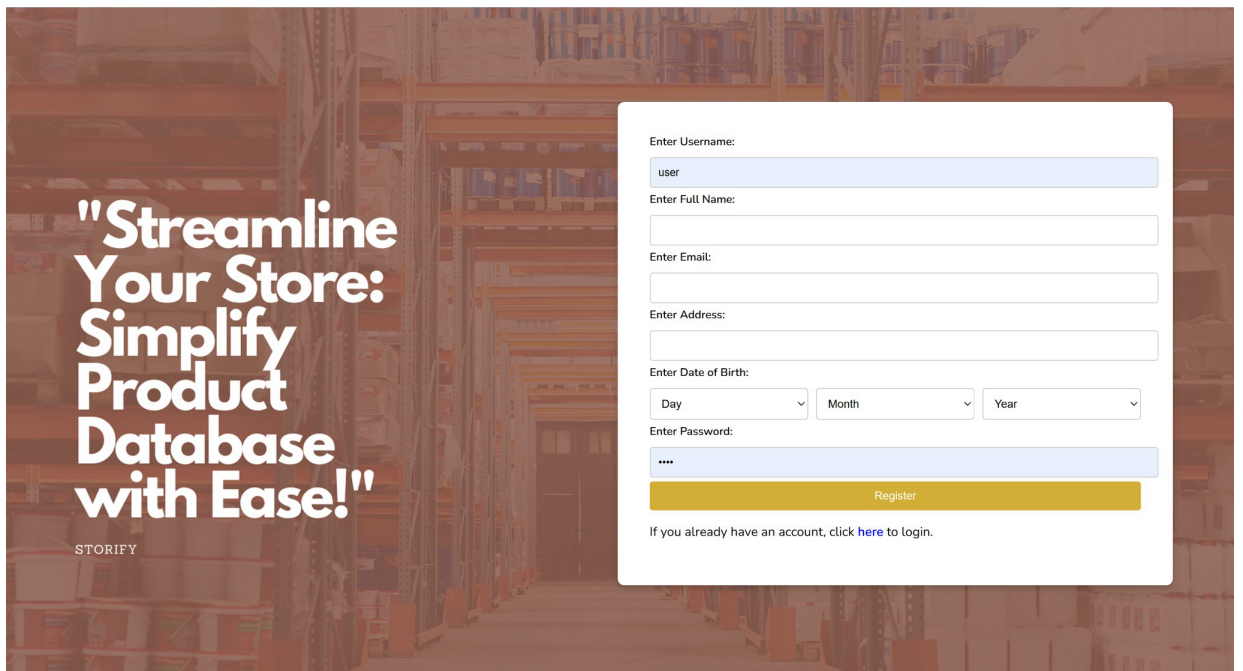


Features and Screenshots

1. Login



2. Register



3. Input Product

Insert Product

Product Name

Product Quantity

Product Price

Submit

4. Update Product

Update Product

Product Name

Product Quantity

Product Price

Update

5. Dashboard (Show Product)

Welcome to Storify - Product List

Product ID	Product Name	Quantity	Price	Actions
2	Gelas Minum Kaca	8	6000.0	<button>Update</button> <button>Delete</button>
3	Meja Besi	500	500000.0	<button>Update</button> <button>Delete</button>

6. Profile

Your Profile

Name

Rafael Jonathan Raja Nicholas Harianja

Password

Email

rafaeljonathan05@gmail.com

Date of Birth

01/02/2005

Address

Jln. Sukolilo 1,
Surabaya, Jawa

Update Profile

Logout

Code Explanation

1. Login

This feature allows users to access the dashboard by entering the correct username and password. Before entering the dashboard and accessing all the features, users need to login. This

works by checking the user input to the database, if the data matches, it will direct the user to the dashboard page. Otherwise, it will return an alert and return to the login page again.

(Front end)

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Login</title>
<link rel="stylesheet" type="text/css" href="css/index.css">
<link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display
=swap" rel="stylesheet">
</head>
<body>

<form action=LoginServlet method=post>
    <table>
        <tr><td>Username: <input type=text name=txtUsername> </td></tr>
        <tr><td>Password: <input type=password name=txtPassword>
</td></tr>
        <tr><td><input type=submit value=Login></td></tr>
    </table>
<p>If you don't have an account, click <a href="register.jsp">here</a> to
register.</p>
</form>
</body>
</html>
```

```
body {
    font-family:'Nunito', sans-serif;
    margin: 0;
    padding: 0;
    background-image: url('../resource/login_bg.png');
    background-size: cover;
    background-position: center;
    background-repeat: no-repeat;
    display: flex;
```

```
    flex-direction: column;
    justify-content: center;
    align-items: center;
    height: 100vh;

img {
    display: block;
    margin-bottom: 20px;
    max-width: 450px;
}

form {
    background-color: white;
    padding: 20px;
    border-radius: 8px;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    width: 300px;
    text-align: center;
}

form::before {
    content: "Login";
    display: block;
    font-size: 30px;
    font-weight: bold;
    margin-bottom: 20px;
}

table {
    margin: 0 auto;
    width: 100%;
}

td {
    text-align: center;
    padding: 5px 0;
}

input[type="text"], input[type="password"] {
    width: calc(100% - 20px);
```

```

padding: 10px;
margin: 10px 0;
border: 1px solid #ccc;
border-radius: 4px;
font-size: 14px;
}

input[type="submit"] {
background-color: #D4AF37;
color: white;
border: none;
padding: 10px 20px;
margin: 10px 5px;
border-radius: 4px;
font-size: 14px;
cursor: pointer;
transition: background-color 0.3s;
}

input[type="submit"]:hover {
background-color: #B8860B;
}

p a {
color: blue;
text-decoration: none;
}

p a:hover {
text-decoration: underline;
}

```

(Back end)

```

import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

```



```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        try {
            response.setContentType("text/html");
            PrintWriter output = response.getWriter();

            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn =
DriverManager.getConnection("jdbc:mysql://localhost:3306/Storify", "root",
"");

            String user = request.getParameter("txtUsername");
            String pass = request.getParameter("txtPassword");

            String query = "SELECT * FROM Users WHERE UserName = ? AND
UserPassword = ?";
            PreparedStatement ps = conn.prepareStatement(query);
            ps.setString(1, user);
            ps.setString(2, pass);
            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("loggedUser", user);

                RequestDispatcher rd =
request.getRequestDispatcher("dashboard.jsp");
                rd.forward(request, response);
            } else {
```

```

        output.println("<p
style='color:red;text-align:center;'>Invalid username or password. Try
again.</p>");

        RequestDispatcher rd =
request.getRequestDispatcher("login.jsp");
        rd.include(request, response);
    }

    rs.close();
    ps.close();
    conn.close();
} catch (Exception e) {
    e.printStackTrace();
}
}
}

```

Explanation

- Add username and email form, and a link to register if the user has not make an account
- Connect to database with sql connector java and use a search query to find a username and password based on the user input, will direct to dashboard if correct and show an alert if incorrect
- Styling with image backgrounds and white box in the center as a box to contain the forms.

2. Register

The register feature allows the user to create a user account. User needs to fill in the username, user's full name, user's address, user's date of birth, and the password to access the account. Register will inject the user's input to the database using SQL connection. After registering, the user can login with the account that has been registered.

(Front end)

```

<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Register</title>
    <link rel="stylesheet" type="text/css" href="css/register.css">

```

```
<link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display
=swap" rel="stylesheet">
</head>
<body>
  <div class="form-container">
    <form action="RegisterServlet" method="post">
      <label for="txtUsername">Enter Username:</label>
      <input type="text" id="txtUsername" name="txtUsername">

      <label for="txtFullName">Enter Full Name:</label>
      <input type="text" id="txtFullName" name="txtFullName">

      <label for="txtEmail">Enter Email:</label>
      <input type="text" id="txtEmail" name="txtEmail">

      <label for="txtAddress">Enter Address:</label>
      <input type="text" id="txtAddress" name="txtAddress">

      <label>Enter Date of Birth:</label>
      <div class="dob-container">
        <select id="dobDay" name="dobDay">
          <option value="" disabled selected>Day</option>
          <% for (int i = 1; i <= 31; i++) { %>
            <option value="<%= i %>"><%= i %></option>
          <% } %>
        </select>
        <select id="dobMonth" name="dobMonth">
          <option value="" disabled selected>Month</option>
          <option value="1">January</option>
          <option value="2">February</option>
          <option value="3">March</option>
          <option value="4">April</option>
          <option value="5">May</option>
          <option value="6">June</option>
          <option value="7">July</option>
          <option value="8">August</option>
          <option value="9">September</option>
          <option value="10">October</option>
          <option value="11">November</option>
        </select>
      </div>
    </form>
  </div>
</body>
</html>
```

```

        <option value="12">December</option>
    </select>
    <select id="dobYear" name="dobYear">
        <option value="" disabled selected>Year</option>
        <% for (int i = 1900; i <= 2024; i++) { %>
            <option value="<%= i %>"><%= i %></option>
        <% } %>
    </select>
</div>

<label for="txtPassword">Enter Password:</label>
<input type="password" id="txtPassword" name="txtPassword">

<input type="submit" value="Register">

<p>If you already have an account, click <a
href="login.jsp">here</a> to login.</p>
</form>
</div>
</body>
</html>

```

```

body {
    font-family: 'Nunito', sans-serif;
    margin: 0;
    padding: 0;
    background-image: url('../resource/register_bg.png');
    background-size: cover;
    background-position: center;
    background-repeat: no-repeat;
    display: flex;
    justify-content: flex-end;
    align-items: center;
    height: 100vh;
}

.form-container {
    background-color: white;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    border-radius: 8px;
}

```

```
padding: 40px;
width: 40%;
margin-right: 5%;
display: flex;
flex-direction: column;
gap: 20px;
}

.form-container input[type="text"],
.form-container input[type="password"],
.form-container select {
  width: 100%;
  padding: 10px;
  margin: 5px 0;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 14px;
}

.form-container label {
  font-weight: bold;
  margin-bottom: 5px;
  display: block;
  font-size: 14px; /

}

.dob-container {
  display: flex;
  gap: 10px;
}

.dob-container select {
  flex: 1;
}

.form-container input[type="submit"] {
  background-color: #D4AF37;
  color: white;
  border: none;
  padding: 10px 20px;
```

```

border-radius: 4px;
font-size: 14px;
cursor: pointer;
transition: background-color 0.3s;
width: 100%;
}

.form-container input[type="submit"]:hover {
background-color: #B8860B;
}

.form-container p a {
color: blue;
text-decoration: none;
font-weight: bold;
}

.form-container p a:hover {
text-decoration: underline;
}

@media (max-width: 768px) {
body {
justify-content: center;
}

.form-container {
width: 90%;
margin-right: 0;
}

.dob-container {
flex-direction: column;
}
}

```

(Back end)

```

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;

```

```
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

@WebServlet("/RegisterServlet")
public class RegisterServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        String url = "jdbc:mysql://localhost:3306/Storify";
        String username = "root";
        String password = "";
        PrintWriter out = response.getWriter();
        response.setContentType("text/html");

        String userName = request.getParameter("txtUsername");
        String fullName = request.getParameter("txtFullName");
        String email = request.getParameter("txtEmail");
        String address = request.getParameter("txtAddress");
        String passwordInput = request.getParameter("txtPassword");

        String dobDay = request.getParameter("dobDay");
        String dobMonth = request.getParameter("dobMonth");
        String dobYear = request.getParameter("dobYear");
        String dateOfBirth = dobYear + "-" + dobMonth + "-" + dobDay;

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, username,
password);
```

```

        String sql = "INSERT INTO Users (UserEmail, UserName,
UserPassword, UserDOB, UserFullName, UserAddress) VALUES (?, ?, ?, ?, ?,
?)" ;

        PreparedStatement ps = conn.prepareStatement(sql);
        ps.setString(1, email);
        ps.setString(2, userName);
        ps.setString(3, passwordInput);
        ps.setString(4, dateOfBirth);
        ps.setString(5, fullName);
        ps.setString(6, address);

        int rows = ps.executeUpdate();

        if (rows > 0) {
            out.println("<script type='text/javascript'>");
            out.println("alert('Registration successful! You can now
log in.');"");
            out.println("window.location.href = 'login.jsp'");
            out.println("</script>");
        } else {
            out.println("<script type='text/javascript'>");
            out.println("alert('Registration failed! Please try
again.');"");
            out.println("window.location.href = 'register.jsp'");
            out.println("</script>");
        }

        conn.close();
    } catch (Exception e) {
        e.printStackTrace();
        out.println("<h1>Error occurred: " + e.getMessage() +
"</h1>");
    }
}
}

```

Explanation:

- Add forms for users to fill with register information which consists of username, user full name, user date of birth, user email, and password.

- Add styling which adds a picture as a background and white box on the right side to contain the forms.
- Using a java sql connector, connect the front end to the database. Insert the user's input to the database by using a query INSERT INTO Products.

3. Input Product

This feature allows users to insert a product into the database. Users can fill in the product's name, product's price and its quantity. This feature works similarly like the register feature, the product ID will be auto generated in order.

(Front end)

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<link rel="stylesheet" type="text/css" href="css/inputproduct.css">
<link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display
=swap" rel="stylesheet">
<title>Storify</title>
</head>
<body>
    <ul>
        <li><a href="dashboard.jsp"></a></li>
        <li><a class="active" href="InputProduct.jsp">Input
Product</a></li>
        <li><a href="Profile.jsp">Profile</a></li>
    </ul>
    <main>
        <div class="form-container">
            <h1>Insert Product</h1>
            <form action="InputProductServlet" method="post">
                <table>
                    <tr>
                        <td><label for="productName">Product
Name</label></td>
```

```

        <td><input type="text" id="productName"
name="productName" required></td>
    </tr>
    <tr>
        <td><label for="productQuantity">Product
Quantity</label></td>
        <td><input type="number" id="productQuantity"
name="productQuantity" required></td>
    </tr>
    <tr>
        <td><label for="productPrice">Product
Price</label></td>
        <td><input type="number" step="0.01"
id="productPrice" name="productPrice" required></td>
    </tr>
    <tr>
        <td colspan="2" style="text-align: center;">
            <button type="submit">Submit</button>
        </td>
    </tr>
</table>
</form>
</div>
</main>
</body>
</html>

```

```

body {
    margin: 0;
    font-family: 'Nunito', sans-serif;
}

/* Navbar styling */
ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    overflow: hidden;
    background-color: #F0DBC5;
    display: flex;

```

```
        height: 55px;
        align-items: center;
    }

    /* Navbar items */
    ul li {
        display: inline;
    }

    /* Links styling */
    ul li a {
        color: black;
        text-decoration: none;
        display: block;
        padding: 25px 25px;
        text-align: center;
    }

    /* Hover effect */
    ul li a:hover {
        background-color: #C4B5A5;
    }

    .active {
        background-color: #C4B5A5;
    }

    /* Alignment: left for first item, right for last item */
    ul li:first-child {
        height: auto;
    }

    ul li:last-child {
        margin-left: auto;
    }

    main {
        padding: 20px;
        max-width: 900px;
        margin: auto;
    }
}
```

```

}

/* Style the form container */
.form-container {
    background-color: white;
    padding: 20px 30px;
    border-radius: 8px;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    margin: 80px 200px;
}

.form-container h1 {
    text-align: center;
    margin-bottom: 20px;
}

.form-container table {
    width: 100%;
    border-collapse: collapse;
}

.form-container tbody td:first-child {
    width: 165px; /* Lebar tetap untuk kolom "No" */
}

.form-container label {
    display: block;
    margin-bottom: 5px;
    font-weight: bold;
}

.form-container input {
    width: calc(100% - 10px);
    padding: 8px;
    border: 1px solid #ccc;
    border-radius: 5px;
    box-sizing: border-box;
    margin-bottom: 10px;
}

.form-container .button-container {

```

```

        text-align: right;
        margin-top: 10px;
    }

    .form-container button {
        padding: 8px 20px;
        font-size: 14px;
        font-weight: bold;
        background-color: #D4AF37;
        color: white;
        border: none;
        border-radius: 5px;
        cursor: pointer;
    }

    .form-container button:hover {
        background-color: #B8860B;
    }

```

(Back end)

```

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

@WebServlet("/InputProductServlet")
public class InputProductServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        String url = "jdbc:mysql://localhost:3306/Storify";
        String username = "root";

```

```

String password = "";
PrintWriter out = response.getWriter();
response.setContentType("text/html");

String name = request.getParameter("productName");
String quantity = request.getParameter("productQuantity");
String price = request.getParameter("productPrice");

try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection conn = DriverManager.getConnection(url, username,
password);

    String sql = "INSERT INTO Products (productName,
productQuantity, productPrice) VALUES (?, ?, ?)";
    PreparedStatement ps = conn.prepareStatement(sql);
    ps.setString(1, name);
    ps.setString(2, quantity);
    ps.setString(3, price);

    int rows = ps.executeUpdate();

    if (rows > 0) {
        out.println("<script type='text/javascript'>");
        out.println("alert('Product Input success!');");
        out.println("window.location.href = 'InputProduct.jsp';");
        out.println("</script>");
    } else {
        out.println("<script type='text/javascript'>");
        out.println("alert('Product Input failed! Please try
again.');

```

```

    }
}
}

```

Explanation:

- Use form to get user input that will later be inserted to the database.
- Create a box to contain the forms and color the buttons
- Add a SQL connector (Java) to connect the user input with the database using SQL Query.

4. Profile

Profile shows the user's current profile. It shows all the user's information on its account except the userID. Since userID is used by us to mark the user.

Frontend :

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.Connection" %>
<%@ page import="java.sql.DriverManager" %>
<%@ page import="java.sql.PreparedStatement" %>
<%@ page import="java.sql.ResultSet" %>
<%@ page import="jakarta.servlet.http.HttpSession" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<link rel="stylesheet" type="text/css" href="css/profile.css">
<link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display
=swap" rel="stylesheet">
<title>Storify</title>
</head>
<body>
    <ul>
        <li><a href="dashboard.jsp"></a></li>
        <li><a href="InputProduct.jsp">Input Product</a></li>
        <li><a class="active" href="profile.jsp">Profile</a></li>
    </ul>
    <main>

```

```

<div class="form-container">
    <h1>Your Profile</h1>
    <form action="UpdateUserServlet" method="post">
        <%
            // Mendapatkan username dari session
            String loggedUser = (String)
session.getAttribute("loggedUser");

            if (loggedUser == null) {
                // Jika tidak ada user yang login, redirect ke halaman
login
                response.sendRedirect("login.jsp");
            } else {
                try {
                    Class.forName("com.mysql.cj.jdbc.Driver");
                    String url =
"jdbc:mysql://localhost:3306/storify";
                    String username = "root";
                    String password = "";
                    String query = "SELECT * FROM users WHERE UserName
= ?";

                    Connection conn = DriverManager.getConnection(url,
username, password);

                    PreparedStatement ps =
conn.prepareStatement(query);
                    ps.setString(1, loggedUser);
                    ResultSet rs = ps.executeQuery();

                    if (rs.next()) {
                        %>
                        <input type="hidden" name="username" value="<%= loggedUser
%>">

                        <table>
                            <tr>
                                <td><label>Name</label></td>
                                <td><input type="text" name="name" value="<%=
rs.getString("UserFullName") %>"></td>
                            </tr>
                            <tr>

```



```

        <td><label>Password</label></td>
        <td><input type="password" name="password"
value="<%= rs.getString("UserPassword") %>"></td>
    </tr>
    <tr>
        <td><label>Email</label></td>
        <td><input type="email" name="email" value="<%=
rs.getString("UserEmail") %>"></td>
    </tr>
    <tr>
        <td><label>Date of Birth</label></td>
        <td><input type="date" name="dob" value="<%=
rs.getDate("UserDOB") %>"></td>
    </tr>
    <tr>
        <td><label>Address</label></td>
        <td><textarea name="address"><%=
rs.getString("UserAddress") %></textarea></td>
    </tr>
    <tr>
        <td colspan="2" style="text-align: center;">
            <button type="submit">Update Profile</button>
        </td>
    </tr>
</table>
<%
    }
    rs.close();
    ps.close();
    conn.close();
} catch (Exception e) {
    e.printStackTrace();
    out.println("<p>Error retrieving user data.</p>");
}
}
%>
</form>

<form action="LogoutServlet" method="post">

```

```
        <button type="submit">Logout</button>
    </form>
</div>
</main>
</body>
</html>
```

```
body {
    margin: 0;
    font-family: 'Nunito', sans-serif;
}
```

```
ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    overflow: hidden;
    background-color: #F0DBC5;
    display: flex;
    height: 55px;
    align-items: center;
}
```

```
ul li {
    display: inline;
}
```

```
ul li a {
    color: black;
    text-decoration: none;
    display: block;
    padding: 25px 25px;
    text-align: center;
}
```

```
ul li a:hover {
```

```
        background-color: #C4B5A5;
    }

    .active {
        background-color: #C4B5A5;
    }

    ul li:first-child {
        height: auto;
    }

    ul li:last-child {
        margin-left: auto;
    }

    main {
        padding: 20px;
        max-width: 900px;
        margin: auto;
    }

    .form-container {
        background-color: white;
        padding: 20px 30px;
        border-radius: 8px;
        box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
        margin: 80px 200px;
    }

    .form-container h1 {
        text-align: center;
        margin-bottom: 20px;
    }

    .form-container table {
        width: 100%;
        border-collapse: collapse;
    }
```

```
.form-container tbody td:first-child {
    width: 165px;
}

.form-container label {
    display: block;
    margin-bottom: 5px;
    font-weight: bold;
}

.form-container input {
    width: calc(100% - 10px);
    padding: 8px;
    border: 1px solid #ccc;
    border-radius: 5px;
    box-sizing: border-box;
    margin-bottom: 10px;
}

.form-container .button-container {
    text-align: right;
    margin-top: 10px;
}

.form-container button {
    padding: 8px 20px;
    font-size: 14px;
    font-weight: bold;
    background-color: #D4AF37;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

.form-container button:hover {
    background-color: #B8860B;
}
```

Explanation :

- Includes several fields that allow the user to update their personal information. The form contains a text field for the user's Full Name, a password field for updating the Password, an email field to modify their Email Address, a date field for updating their Date of Birth, and a text area for the user to input or edit their Address.
- There will be a logout button below the box.

5. Update Profile

This will enable the user to view and change their information at the same time, if they want to view only, then they don't need to click the update profile button. If they want to change, then type a new variable and then click update profile to update it.

(Front end)

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<link rel="stylesheet" type="text/css" href="css/inputproduct.css">
<link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display=swap" rel="stylesheet">
<title>Update User</title>
</head>
<body>
    <ul>
        <li><a href="dashboard.jsp"></a></li>
        <li><a href="Profile.jsp">Profile</a></li>
    </ul>
    <main>
        <div class="form-container">
            <h1>Update Profile</h1>
            <form action="UpdateUserServlet" method="post">
                <input type="hidden" name="userId" value="${user.UserId}">
                <table>
                    <tr>
                        <td><label for="userName">Username</label></td>
```

```

        <td><input type="text" id="userName"
name="userName" value="${user.UserName}" required></td>
    </tr>
    <tr>
        <td><label for="userFullName">Full
Name</label></td>
        <td><input type="text" id="userFullName"
name="userFullName" value="${user.UserFullName}" required></td>
    </tr>
    <tr>
        <td><label for="userEmail">Email</label></td>
        <td><input type="email" id="userEmail"
name="userEmail" value="${user.UserEmail}" required></td>
    </tr>
    <tr>
        <td><label for="userAddress">Address</label></td>
        <td><input type="text" id="userAddress"
name="userAddress" value="${user.UserAddress}" required></td>
    </tr>
    <tr>
        <td colspan="2" style="text-align: center;">
            <button type="submit">Update</button>
        </td>
    </tr>
</table>
</form>
</div>
</main>
</body>
</html>

```

(Back end)

```

import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;

```

```

import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/UpdateUserServlet")
public class UpdateUserServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        String loggedUser = (session != null) ? (String)
session.getAttribute("loggedUser") : null;

        if (loggedUser == null) {
            response.getWriter().println("<p>You are not authorized to
update this profile.</p>");
            return;
        }

        String name = request.getParameter("name");
        String password = request.getParameter("password");
        String email = request.getParameter("email");
        String dob = request.getParameter("dob");
        String address = request.getParameter("address");

        try (Connection conn =
DriverManager.getConnection("jdbc:mysql://localhost:3306/storify", "root",
"")) {
            PreparedStatement pstmt = conn.prepareStatement(
                "UPDATE users SET UserFullName=?, UserPassword=?,
UserEmail=?, UserDOB=?, UserAddress=? WHERE UserName=?"
            ) {
                pstmt.setString(1, name);
                pstmt.setString(2, password);
                pstmt.setString(3, email);
                pstmt.setString(4, dob);
                pstmt.setString(5, address);
                pstmt.setString(6, loggedUser);
            }
        }
    }
}

```

```

        int updatedRows = pstmt.executeUpdate();
        if (updatedRows > 0) {
            response.sendRedirect("profile.jsp");
        } else {
            response.getWriter().println("<p>Failed to update profile.
Try again later.</p>");
        }
    } catch (Exception e) {
        e.printStackTrace();
        response.getWriter().println("<p>Error updating
profile.</p>");
    }
}
}

```

Explanation :

- Checks the session to ensure the user is logged in and retrieves the updated data from the form.
- Connects to the MySQL database and executes an SQL query to update the user's profile information (name, password, email, date of birth, and address) using a PreparedStatement.
- If the update is successful, the user is redirected to profile.jsp; otherwise, an error message is displayed.

6. Dashboard

This is the main interface for everything, this will display all products that exist in the product database. Users can click edit which will be directed to edit product page, or delete, which will delete the product from the database. Before deleting, there will be an alert before deleting to prevent misclick.

(Front end)

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8" %>
<%@ page import="java.sql.Connection" %>
<%@ page import="java.sql.DriverManager" %>
<%@ page import="java.sql.ResultSet" %>
<%@ page import="java.sql.Statement" %>

```



```

<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <link rel="stylesheet" type="text/css" href="css/dashboard.css">
    <link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display
=swap" rel="stylesheet">
    <title>Storify - Product List</title>
</head>
<body>

<ul>
    <li><a class="active" href="dashboard.jsp"></a></li>
    <li><a href="InputProduct.jsp">Input Product</a></li>
    <li><a href="profile.jsp">Profile</a></li>
</ul>

<header>
    <h1>Welcome to Storify - Product List</h1>
</header>

<main>
    <table border="1">
        <thead>
            <tr>
                <th>Product ID</th>
                <th>Product Name</th>
                <th>Quantity</th>
                <th>Price</th>
                <th>Actions</th>
            </tr>
        </thead>
        <tbody>
            <%
try {
    Class.forName("com.mysql.jdbc.Driver");
    String url = "jdbc:mysql://localhost:3306/storify";
    String username = "root";

```

```

String password = "";
String query = "SELECT * FROM products";

Connection conn = DriverManager.getConnection(url, username,
password);

Statement stmt = conn.createStatement();
ResultSet rs = stmt.executeQuery(query);

while (rs.next()) {
    int productId = rs.getInt("ProductId");
    %>
    <tr>
        <td><%= productId %></td>
        <td><%= rs.getString("ProductName") %></td>
        <td><%= rs.getInt("ProductQuantity") %></td>
        <td><%= rs.getDouble("ProductPrice") %></td>
        <td class="action-buttons">

            <form action="UpdateProductServlet" method="GET"
style="display: inline;">
                <input type="hidden" name="productId" value="<%=
productId %>">
                <button type="submit" class="btn
update-btn">Update</button>
            </form>

            <form action="DeleteProductServlet" method="POST"
style="display: inline;"
                onsubmit="return confirm('Are you sure you want
to delete this product?');">
                <input type="hidden" name="productID" value="<%=
productId %>">
                <input type="hidden" name="productName" value="<%=
rs.getString("ProductName") %>">
                <button type="submit" class="btn
delete-btn">Delete</button>
            </form>
        </td>
    </tr>

```

```

        <%
            }
            rs.close();
            stmt.close();
            conn.close();
        } catch (Exception e) {
            e.printStackTrace();
            out.println("<tr><td colspan='5'>Error retrieving
products.</td></tr>");
        }
    %>
</tbody>
</table>
</main>
</body>
</html>

```

```

body {
    margin: 0;
    font-family: 'Nunito', sans-serif;
}

ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    overflow: hidden;
    background-color: #F0DBC5;
    display: flex;
    height: 55px;
    align-items: center;
}

ul li {
    display: inline;
}

```

```
ul li a {
    color: black;
    text-decoration: none;
    display: block;
    padding: 25px 25px;
    text-align: center;
}

ul li a:hover {
    background-color: #C4B5A5;
}

.active {
    background-color: #C4B5A5;
}

ul li:first-child {
    height: auto;
}

ul li:last-child {
    margin-left: auto;
}

header {
    text-align: center;
    padding: 20px;
    color: black;
}

header h1 {
    margin: 0;
}

main {
    padding: 20px;
```

```
    max-width: 900px;
    margin: auto;
}

main h1 {
    text-align: center;
    margin-bottom: 20px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
    background: white;
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.5);
}

thead th {
    background-color: #F0DBC5;
    color: black;
    padding: 10px;
    text-align: left;
    font-weight: 600;
    border: 1px solid black;
}

tbody td {
    padding: 10px;
    border: 1px solid black;
}

thead th:first-child, tbody td:first-child {
    width: 30px;
    text-align: center;
}

table, th, td, tr {
```

```

        border: none;
    }

    .delete-btn {
        background-color: #ff4444;
        color: white;
        border: none;
        padding: 5px 10px;
        border-radius: 3px;
        cursor: pointer;
    }

    .delete-btn:hover {
        background-color: #cc0000;
    }

    .update-btn {
        background-color: blue;
        color: white;
        border: none;
        padding: 5px 10px;
        border-radius: 3px;
        cursor: pointer;
    }

    .update-btn:hover {
        background-color: darkblue;
    }

```

Explanation :

- Table that will get the product database.
- Style it to create a table with a specific color.
- Will GET the data from the database using SELECT all products in the database.
- Add 2 buttons (action) delete and update that will direct to update product/run the command delete product on the selected row.

7. UpdateProduct :

This will allow the user to edit the selected products name, price or quantity. User just need to input the new value and press the button to update.

(Front end)

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>
<%@ page import="java.sql.ResultSet" %>
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <link rel="stylesheet" type="text/css" href="css/inputproduct.css">
    <link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600&display
=swap" rel="stylesheet">
    <title>Update Product</title>
    <style>
        .form-container {
            max-width: 500px;
            margin: 20px auto;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
        }

        button[type="submit"] {
            background-color: #4CAF50;
            color: white;
            padding: 10px 20px;
            border: none;
            border-radius: 4px;
            cursor: pointer;
            font-size: 16px;
        }

        button[type="submit"]:hover {
            background-color: #45a049;
        }

        input[type="text"],
        input[type="number"] {
            width: 100%;
```

```

        padding: 8px;
        margin: 5px 0;
        border: 1px solid #ddd;
        border-radius: 4px;
        box-sizing: border-box;
    }

    table {
        width: 100%;
    }

    td {
        padding: 10px;
    }

    label {
        font-weight: bold;
    }
</style>
</head>
<body>
    <ul>
        <li><a href="dashboard.jsp"></a></li>
        <li><a href="InputProduct.jsp">Input Product</a></li>
        <li><a href="Profile.jsp">Profile</a></li>
    </ul>

    <main>
        <div class="form-container">
            <h1>Update Product</h1>
            <%
                ResultSet rs = (ResultSet)request.getAttribute("product");
                if(rs != null) {
            %>
            <form action="UpdateProductServlet" method="post">
                <input type="hidden" name="productId" value="<%=
rs.getInt("ProductId") %>">
                <table>
                    <tr>

```



```

                <td><label for="productName">Product
Name</label></td>
                <td><input type="text" id="productName"
name="productName"
                value="<%= rs.getString("ProductName")
%>" required></td>
            </tr>
            <tr>
                <td><label for="productQuantity">Product
Quantity</label></td>
                <td><input type="number" id="productQuantity"
name="productQuantity"
                value="<%= rs.getInt("ProductQuantity")
%>" min="0" required></td>
            </tr>
            <tr>
                <td><label for="productPrice">Product
Price</label></td>
                <td><input type="number" step="0.01"
id="productPrice" name="productPrice"
                value="<%= rs.getDouble("ProductPrice")
%>" min="0" required></td>
            </tr>
            <tr>
                <td colspan="2" style="text-align: center;">
                    <button type="submit">Update</button>
                </td>
            </tr>
        </table>
    </form>
    <%
        } else {
    %>
        <p style="color: red; text-align: center;">Error: Product
data not found!</p>
        <p style="text-align: center;">
            <a href="dashboard.jsp" style="color: blue;
text-decoration: none;">Return to Dashboard</a>
        </p>
    <%

```

```

        }

        %>
    </div>
</main>
</body>
</html>

```

(Back end)

```

import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

@WebServlet("/UpdateProductServlet")
public class UpdateProductServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;
    private static final String DB_URL =
"jdbc:mysql://localhost:3306/Storify";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "";

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        try {
            int productId =
Integer.parseInt(request.getParameter("productId"));

```

```

        Class.forName("com.mysql.cj.jdbc.Driver");

        try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD)) {
            PreparedStatement ps = conn.prepareStatement("SELECT *
FROM Products WHERE ProductId = ?");

            ps.setInt(1, productId);
            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                // Store product details in request attributes
                request.setAttribute("product", rs);
                RequestDispatcher rd =
request.getRequestDispatcher("UpdateProduct.jsp");
                rd.forward(request, response);
            } else {
                out.println("<script type='text/javascript'>");
                out.println("alert('Product not found!');");
                out.println("window.location='dashboard.jsp';");
                out.println("</script>");
            }
        }
    } catch (NumberFormatException e) {
        out.println("<script type='text/javascript'>");
        out.println("alert('Invalid product ID!');");
        out.println("window.location='dashboard.jsp';");
        out.println("</script>");
    } catch (ClassNotFoundException | SQLException e) {
        out.println("<script type='text/javascript'>");
        out.println("alert('Database error: " + e.getMessage() +
"');");
        out.println("window.location='dashboard.jsp';");
        out.println("</script>");
    }
}

protected void doPost(HttpServletRequest request, HttpServletResponse
response)
    throws ServletException, IOException {

```

```

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        try {
            // Validate and parse input
            int productId =
Integer.parseInt(request.getParameter("productId"));
            String productName = request.getParameter("productName");
            int productQuantity =
Integer.parseInt(request.getParameter("productQuantity"));
            double productPrice =
Double.parseDouble(request.getParameter("productPrice"));

            // Basic validation
            if (productName == null || productName.trim().isEmpty()) {
                throw new IllegalArgumentException("Product name cannot be
empty");
            }
            if (productQuantity < 0) {
                throw new IllegalArgumentException("Product quantity
cannot be negative");
            }
            if (productPrice < 0) {
                throw new IllegalArgumentException("Product price cannot
be negative");
            }

            Class.forName("com.mysql.cj.jdbc.Driver");
            try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD)) {
                PreparedStatement ps = conn.prepareStatement(
                    "UPDATE Products SET ProductName = ?, ProductQuantity
= ?, ProductPrice = ? WHERE ProductId = ?"
                );
                ps.setString(1, productName);
                ps.setInt(2, productQuantity);
                ps.setDouble(3, productPrice);
                ps.setInt(4, productId);

                int result = ps.executeUpdate();
            }
        }
    }
}

```

```

        if (result > 0) {
            out.println("<script type='text/javascript'>");
            out.println("alert('Product updated successfully!');");
            out.println("window.location='dashboard.jsp';");
            out.println("</script>");
        } else {
            out.println("<script type='text/javascript'>");
            out.println("alert('Failed to update product!');");
            out.println("window.location='dashboard.jsp';");
            out.println("</script>");
        }
    }
} catch (NumberFormatException e) {
    out.println("<script type='text/javascript'>");
    out.println("alert('Invalid number format in input!');");
    out.println("window.location='dashboard.jsp';");
    out.println("</script>");
} catch (IllegalArgumentException e) {
    out.println("<script type='text/javascript'>");
    out.println("alert('" + e.getMessage() + "');");
    out.println("window.location='dashboard.jsp';");
    out.println("</script>");
} catch (ClassNotFoundException | SQLException e) {
    out.println("<script type='text/javascript'>");
    out.println("alert('Database error: " + e.getMessage() +
"'');");
    out.println("window.location='dashboard.jsp';");
    out.println("</script>");
}
}
}

```

- Add a form that allows users to fill a new value for the product.
- Style it to create a form inside the white box.
- Connect to the SQL Database and will send and UPDATE query of the selected product ID and will update product's value except the ID

8. Logout :

Front end is located in the profile's page. It will act as a button that will later log the user out and redirect back to the login page.

(Back end)

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import java.io.IOException;

@WebServlet("/LogoutServlet")
public class LogoutServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session != null) {
            session.invalidate();
        }
        response.sendRedirect("login.jsp");
    }
}
```

Explanation :

- POST request is made to this servlet, it first checks if the user has an active session by calling request.getSession(false).
- If the session exists, it invalidates the session using session.invalidate(), which effectively logs the user out by removing any session data.
- After invalidating the session, the servlet redirects the user to the login.jsp page, prompting them to log in again if they wish to continue using the application.

9. Delete Product

This feature allows the user to delete a product by just pressing a button. This front end is located in the dashboard and shows as a button, and if the button is pressed, it will run this servlet. It will show a warning first before deleting.

(Back end)

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.SQLException;

@WebServlet("/DeleteProductServlet")
public class DeleteProductServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    private static final String DB_URL =
"jdbc:mysql://localhost:3306/Storify";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "";

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter output = response.getWriter();

        String id = request.getParameter("productID");

        Connection conn = null;
        PreparedStatement ps = null;
```

```

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            conn = DriverManager.getConnection(DB_URL, DB_USER,
DB_PASSWORD);

            ps = conn.prepareStatement("DELETE FROM Products WHERE
ProductID = ?");
            ps.setString(1, id);

            int rowsAffected = ps.executeUpdate();

            if(rowsAffected > 0) {
                output.println("<script type='text/javascript'>");
                output.println("alert('Product has been deleted
successfully');");
                output.println("window.location='dashboard.jsp';");
                output.println("</script>");
            } else {
                output.println("<script type='text/javascript'>");
                output.println("alert('Failed to delete product. Product
not found.');");
                output.println("window.location='dashboard.jsp';");
                output.println("</script>");
            }

        } catch (ClassNotFoundException e) {
            output.println("<script type='text/javascript'>");
            output.println("alert('Database driver not found: " +
e.getMessage() + "');");
            output.println("window.location='dashboard.jsp';");
            output.println("</script>");
        } catch (SQLException e) {
            output.println("<script type='text/javascript'>");
            output.println("alert('Database error: " + e.getMessage() +
"');");
            output.println("window.location='dashboard.jsp';");
            output.println("</script>");
        } finally {
            try {
                if (ps != null) ps.close();
            }
        }
    }
}

```



```
        if (conn != null) conn.close();
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
}
```

Explanation :

- Retrieve parameter (row) ID of the product, which identifies which product will be deleted.
- Use DELETE SQL query to delete the selected product.
- If a product is found, it will execute the query, else it will display a deletion failed warning/message.

Contribution

1. Juang Maulana Taruna Putra (5025231257) (33.333%)
 - Register User
 - Dashboard (display product)
 - Login User
 - Creating the report
2. Ahmad Muqorrobin (5025231254) (33.333%)
 - Insert Product
 - Update Product
 - Delete Product
 - Creating the report
3. Kemal Tangguh Aji Rajasa (5025231263) (33.333%)
 - Profile
 - Update profile
 - Log out
 - Creating the report